



International Phonetic Alphabet

The **International Phonetic Alphabet (IPA)** is an alphabetic system of phonetic notation based primarily on the Latin script. It was devised by the International Phonetic Association in the late 19th century as a standard written representation for the sounds of speech.^[1] The IPA is used by linguists, lexicographers, foreign language students and teachers, speech-language pathologists, singers, actors, constructed language creators, and translators.^{[2][3]}

The IPA is designed to represent those qualities of speech that are part of lexical (and, to a limited extent, prosodic) sounds in spoken (oral) language: phones, intonation and the separation of syllables.^[1] To represent additional qualities of speech – such as tooth gnashing, lisping, and sounds made with a cleft palate – an extended set of symbols may be used.^[2]

Segments are transcribed by one or more IPA symbols of two basic types: letters and diacritics. For example, the sound of the English letter <t> may be transcribed in IPA with a single letter: [t], or with a letter plus diacritics: [t̪], depending on how precise one wishes to be. Similarly, the French letter <t> may be transcribed as either [t] or [t̪]: [t̪] and [t̪] are two different, though similar, sounds. Slashes are used to signal phonemic transcription; therefore, /t/ is more abstract than either [t̪] or [t̪] and might refer to either, depending on the context and language.^[note 1]

Occasionally, letters or diacritics are added, removed, or modified by the International Phonetic Association. As of the most recent change in 2005,^[4] there are 107 segmental letters, an indefinitely large number of suprasegmental letters, 44 diacritics (not counting composites), and four extra-lexical prosodic marks in the IPA. These are illustrated in the current IPA chart, posted below in this article and on the International Phonetic Association's website.^[5]

History

In 1886 a group of French and English language teachers, led by the French linguist Paul Passy, formed what would be known from 1897 onwards as the International Phonetic Association (in French, *l'Association phonétique internationale*).^[6] The idea of the alphabet had been suggested to Passy by Otto Jespersen. It was developed by Passy along with other members of the association, principally Daniel Jones. The original IPA alphabet was

International Phonetic Alphabet

[aᵻ pʰiː eᵻ]

"IPA", transcribed narrowly according to Received Pronunciation as [aᵻ pʰiː eᵻ]

Script type	Alphabet – partially <u>featural</u>
Period	1888–present
Languages	Used for phonetic and phonemic transcription of any oral language

Related scripts

Parent systems	<u>Egyptian hieroglyphs</u> <u>Proto-Sinaitic script</u> <u>Phoenician alphabet</u> <u>Greek alphabet</u> <u>Latin script</u> <u>Lepsius Standard Alphabet</u> <u>Palaeotype alphabet</u>, <u>English Phonotypic Alphabet</u> <u>Romic alphabet</u> <u>International Phonetic Alphabet</u>
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Unicode

Unicode range	<i>See <u>Phonetic symbols in Unicode § Unicode blocks</u></i>
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based on the Romic alphabet, an English spelling reform created by Henry Sweet that in turn was based on the Palaeotype alphabet of Alexander John Ellis, itself derived from Lepsius Standard Alphabet first used for transcribing Ancient Egyptian into German.

The original intent was to make it usable for other languages the values of the symbols were allowed to vary from language to language.^[note 2] For example, the sound [ʃ] (the *sh* in *shoe*) was originally represented with the letter ⟨c⟩ for English but with ⟨x⟩ for French and German; with German, ⟨c⟩ was used for the [x] sound of *Bach*.^[6] With a growing number of transcribed languages this proved impractical, and in 1888 the values of the letters were made uniform across languages. This would provide the base for all future revisions.^{[6][8]}

Since its creation, the IPA has undergone a number of revisions. After relatively frequent revisions and expansions from the 1890s to the 1940s, the IPA remained nearly static until the Kiel Convention in 1989, which substantially revamped the alphabet. A smaller revision took place in 1993 with the resurrection of letters for mid central vowels^[2] and the retirement of letters for voiceless implosives.^[9] The alphabet was last revised in May 2005 with the addition of a letter for a labiodental flap.^[10] Apart from the addition and removal of symbols, changes to the IPA have consisted largely of renaming symbols and categories and in modifying typefaces.^[2]

Extensions to the International Phonetic Alphabet for speech pathology (extIPA) were created in 1990 and were officially adopted by the International Clinical Phonetics and Linguistics Association in 1994.^[11] They were substantially revised in 2015 with lesser changes in 2025.

Description

The general principle of the IPA is to provide one letter for each distinctive sound (phoneme).^[note 3] This means that:

- It does not use combinations of letters to represent single sounds, the way English does with ⟨sh⟩ and ⟨ea⟩, nor single letters to represent multiple sounds, the way ⟨x⟩ represents /ks/ or /gz/ in English.^[note 4]
- There are no letters that have context-dependent sound values, the way English ⟨c⟩ and ⟨g⟩ have a "hard" and "soft" pronunciation.
- The IPA does not generally have separate letters for two sounds if no known language makes a distinction between them, a property known as "selectiveness".^{[2][note 5]} However, if a large number of phonemically distinct letters can be derived with a single diacritic, that may be used instead.^[note 6]

The alphabet is designed for transcribing sounds (phones), not phonemes, though it is used for phonemic transcription as well. A few letters that did not indicate specific sounds have been retired – ⟨˘⟩, once used for the "compound" tone of Swedish and Norwegian, and ⟨ŋ⟩, once used for the moraic nasal of Japanese – though one remains: ⟨j⟩, used for the *sj*-sound of Swedish. When the IPA is used for broad phonetic or for phonemic transcription, the letter–sound correspondence can be rather loose. The IPA has recommended that more 'familiar' letters be used when that would not cause ambiguity.^[13] For example, ⟨e⟩ and ⟨o⟩ for [ɛ] and [ɔ], ⟨t⟩ for [t̪] or [t̺], ⟨f⟩ for [ɸ], etc. Indeed, in the illustration of Hindi in the *IPA Handbook*, the letters ⟨c⟩ and ⟨ɟ⟩ are used for /t̪ʃ/ and /d̪ʒ/.

Among the symbols of the IPA, 107 letters represent consonants and vowels, 31 diacritics are used to modify these, and 17 additional signs indicate suprasegmental qualities such as length, tone, stress, and intonation.^[note 7] These are organized into a chart; the chart displayed here is the official chart as posted at the website of the IPA.

Greek form, while for ⟨β ~ β⟩ and ⟨χ ~ χ⟩, both Greek and Latin forms are in common use.^[16]



An example of a font that uses turned small-capital omega ⟨ʊ⟩ for the vowel letter ⟨ʊ⟩. The glyph had originally been a small-capital ⟨u⟩.

Iconicity

The graphic derivation of letters and diacritics may be iconic

- A rightward-facing hooked tail, as in ⟨t̪ d̪ ɳ ɾ ʂ ʐ ɻ ɭ⟩, indicates retroflex articulation. It originates from the hooked arm of an *r*.
- The top hook, as in ⟨ɡ d̪ b̪⟩, indicates implosion.
- Several nasal consonants are based on the form ⟨n⟩: ⟨n ɲ ŋ ɳ⟩. ⟨ɲ⟩ and ⟨ŋ⟩ derive from ligatures of *gn* and *ng*, and ⟨ɳ⟩ is an *ad hoc* imitation of ⟨ŋ⟩.
- Among consonant letters, the small capital letters ⟨ɠ ɥ ɮ ɳ ɾ ʂ ʐ ɻ ɭ⟩, and also ⟨ɣ⟩ in extIPA, indicate more guttural sounds than their base letters – ⟨ɸ⟩ is a late exception. Among vowel letters, the small capitals ⟨ɪ ʏ ʊ⟩ indicate what had originally been considered more lax articulations than their base letters; ⟨ʊ⟩ had originally been ⟨u⟩.^[17] Again, small-cap ⟨œ⟩ is a late exception.
- The tone letters derive from a pitch trace on a musical scale.

Brackets and transcription delimiters

There are two principal types of brackets used to set off (delimit) IPA transcriptions:

Symbol	Use
[...]	Square brackets are used with phonetic notation, whether broad or narrow ^[18] – that is, for actual pronunciation, possibly including details of the pronunciation that may not be used for distinguishing words in the language being transcribed, but which the author nonetheless wishes to document. Such phonetic notation is the primary function of the IPA.
/ ... /	Slashes ^[note 11] are used for abstract phonemic notation, ^[18] which note only features that are distinctive in the language, without any extraneous detail. For example, while the 'p' sounds of English <i>pin</i> and <i>spin</i> are pronounced differently (and this difference would be meaningful in some languages), the difference is not meaningful in English. Thus, <i>phonemically</i> the words are usually analyzed as /pɪn/ and /spɪn/, with the same phoneme /p/. To capture the difference between them – the <u>allophones</u> of /p/ – they can be transcribed phonetically as [pʰɪn] and [spɪn]. Phonemic notation commonly uses IPA symbols that are rather close to the default pronunciation of a phoneme, but for legibility often uses simple and 'familiar' letters rather than precise notation, for example /r/ and /o/ for the English [ɹ] and [əʊ] sounds, or /c, ɹ/ for [tʃ, dʒ] as mentioned above.

Less common conventions include:

Symbol	Use
{ ... }	Braces ("curly brackets") are used for <u>prosodic</u> notation. ^[19] See <u>Extensions to the International Phonetic Alphabet</u> for examples in this system.
(...)	Parentheses are used for indistinguishable^[18] or unidentified utterances. They are also seen for silent articulation (mouthing),^[20] where the expected phonetic transcription is derived from lip-reading, and with periods to indicate silent pauses, for example (...) or (2 sec). The latter usage is made official in the <u>extIPA</u>, with unidentified segments circled instead.^[21] In inventory charts, parentheses may indicate that a sound is a marginal phoneme (occurs in only a few morphemes), is only found in loanwords, or is not a phoneme but a notable allophone.
((...))	<u>Double parentheses</u> indicate either a transcription of obscured speech or a description of the obscuring noise. The IPA specifies that they mark the obscured sound,^[19] as in ((2σ)), two audible syllables obscured by another sound. The current extIPA specifications prescribe double parentheses for the extraneous noise, such as ((cough)) for a cough by another person (not the speaker) or ((knock)) for a knock on a door, but the IPA <i>Handbook</i> identifies IPA and extIPA usage as equivalent.^[22] Early publications of the extIPA explain double parentheses as marking "uncertainty because of noise which obscures the recording", and that within them "may be indicated as much detail as the transcriber can detect."^[23]

All three of the above are provided by the IPA *Handbook*. The following are not, but may be seen in IPA transcription or in associated material (especially angle brackets):

Symbol	Field	Description
[[...]]	Phonetics	Double square brackets are used for especially precise phonetic transcription, often finer than is normally practicable. ^[24] This is consistent with the IPA convention of doubling a symbol to indicate greater degree. Double brackets may indicate that a letter has its cardinal IPA value. For example, [a] is an open front vowel, rather than the perhaps slightly different value (such as open central) that "[a]" may be used to transcribe in a particular language. Thus, two vowels transcribed for easy legibility as [e] and [ɛ] may be clarified as actually being [ē] and [ē̃]; [ð] may be more precisely [ð̞]. ^[25] Double brackets may also be used for a specific token or speaker; for example, the pronunciation of a particular child as opposed to the adult pronunciation that is their target. ^[26]
/ ... / { ... } 	Morphophonology	Double slashes are used for morphophonemic transcription. This is also consistent with the IPA convention of doubling a symbol to indicate greater degree – in this case, more abstract than phonemic transcription. Also commonly seen are the braces of <u>set theory</u>, especially when enclosing the set of phonemes that constitute the morphophoneme, e.g. {t d} or {tld} or {/t/, /d/} for a conflated /t/ and /d/. Braces have a conflicting use to delimit prosodic transcription within the <u>Voice Quality Symbols</u>, which are an extension of IPA used in extIPA, but are not otherwise used in IPA proper. Other delimiters sometimes seen are <u>pipes</u> and double pipes taken from <u>Americanist phonetic notation</u>. However, these conflict with the pipes used in basic IPA prosodic transcription.^[note 12]
\ ... \ / ... / ... ! ... !	Diaphonology	Backslashes are used for diaphonemic transcription, for example setting off pronunciations in dictionaries that do not target a specific preferred dialect.^[note 11] Other delimiters are double slashes, – the same notation as for morphophonology, – exclamation marks, and pipes.
< ... > < ... > ...	Graphemics	Angle brackets^[note 13] are used to mark both original Latin orthography and transliteration from another script; they are also used to identify individual <u>graphemes</u> of any script.^{[29][30]} In IPA literature, they are used to indicate the IPA letters themselves rather than the sound values that they carry. For example, <cot> would be used for the orthography of the English word <i>cot</i>, as opposed to its pronunciation /kɒt/. Italics are usual when words are written as themselves (as with <i>cot</i> in the previous sentence) rather than to specifically note their orthography. However, italics are sometimes ambiguous, and italic markup is not always accessible to sight-impaired readers who rely on <u>screen reader</u> technology. Double angle brackets may occasionally be useful to distinguish original orthography from transliteration, or the idiosyncratic spelling of a manuscript from the normalized orthography of the language. Pipes are sometimes used instead of double angle brackets to denote the distinct <u>allographs</u> of a grapheme that are known as <u>glyphs</u>. For example, print g and script g are two glyph variants of the grapheme <g> of Latin script.^[31]

Some examples of contrasting brackets in the literature:

In some English accents, the phoneme /l/, which is usually spelled as <l> or <ll>, is articulated as two distinct allophones: the clear [l] occurs before vowels and the consonant /j/, whereas the dark [ɫ]/[ɭ] occurs before consonants, except /j/, and at the end of words.^[32]

transcribed are necessarily meaningful in the language. The term 'broad' may furthermore carry implication that diacritics are avoided (at least as far as possible) or even that the transcription is restricted to the letters of the ISO basic Latin alphabet.^[47]

For example, the English word *little* may be transcribed broadly as [ˈlɪtəl], approximately describing many pronunciations. A narrower transcription may focus on individual or dialectal details: [ˈhɪɾ] in General American, [ˈlɪɹo] in Cockney, or [ˈhɪː] in Southern US English.

Phonemic transcriptions, which express the conceptual counterparts of spoken sounds, are usually enclosed in slashes (/ /) and tend to use simpler letters with few diacritics. The choice of IPA letters may reflect theoretical claims of how speakers conceptualize sounds as phonemes or they may be merely a convenience for typesetting. Phonemic approximations between slashes do not have absolute sound values. For instance, in English, either the vowel of *pick* or the vowel of *peak* may be transcribed as /i/, so that *pick*, *peak* would be transcribed as /ˈpɪk, ˈpiːk/ or as /ˈpɪk, ˈpɪk/; and neither is identical to the vowel of the French *pique*, which would also be transcribed /pik/. By contrast, a narrow phonetic transcription of *pick*, *peak*, *pique* could be: [pʰɪk], [pʰiːk], [pɪk].

Linguists

IPA is popular for transcription by linguists. Some American linguists, however, use a mix of IPA with Americanist phonetic notation or Sinological phonetic notation or otherwise use nonstandard symbols for various reasons.^[48] Authors who employ such nonstandard use are encouraged to include a chart or other explanation of their choices, which is good practice in general, as linguists differ in their understanding of the exact meaning of IPA symbols and common conventions change over time.

Dictionaries

English

Many British dictionaries, including the *Oxford English Dictionary* and some learner's dictionaries such as the *Oxford Advanced Learner's Dictionary* and the *Cambridge Advanced Learner's Dictionary*, now use the International Phonetic Alphabet to represent the pronunciation of words.^[49] However, most American (and some British) volumes use one of a variety of pronunciation respelling systems, intended to be more

the international phonetic alphabet (2005)

consonants (pulmonic)

	LABIAL		CORONAL				DORSAL			LARYNGEAL		
	Bilabial	Labio-dental	Dental	Alveolar	Palato-alveolar	Retroflex	Alveolo-palatal	Palatal	Velar	Uvular	Pharyngeal	Glottal
Nasal	m	ɱ		n	ɳ		ɲ	ɰ	ŋ			
Plosive	p b			t d	ʈ ɖ		c ɟ	k ɡ	q ɢ		ʔ	ʕ
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ		ħ	ʕ
Approximant		ʋ		ɹ			j		ɰ			
Tap, flap		ɹ̥		ɾ								
Trill	B			r						R	ʜ	ʕ
Lateral fricative				ɬ ɮ								
Lateral approximant				l			ʎ		L			
Lateral flap				ɭ								

consonants (non-pulmonic)

	clicks	implosives	ejectives
ɠ	Bilabial fricated	ɓ	Bilabial
ɠ̊	Laminal alveolar fricated ("dental")	ɗ	Dental or alveolar
ɠ̊̊	Apical (postalveolar) abrupt ("retroflex")	ɖ̥	Retroflex
ɠ̊̊̊	Laminal postalveolar abrupt ("palatal")	ɗ̥	Dental or alveolar
ɠ̊̊̊̊	Lateral alveolar fricated ("lateral")	ɗ̥	Palatal
ɠ̊̊̊̊̊	Velar (back released)	ɗ̥	Velar
		ɗ̥	Lateral alveolar fricative

consonants (co-articulated)

ɱ	Voiced labialized velar approximant	ɰ̥	voiceless labialized velar approximant
ɰ	Voiced labialized velar approximant	ɰ̥	voiceless labialized velar approximant
ɰ̥	Simultaneous x and ɣ (existence disputed)		
ɰ̥̥	Affricates and double articulations may be joined by a tie bar		

brackets

⌈	phonetic/
⌋	phonetic/
⌈	orthographic/
⌋	orthographic/

vowels

suprasegmentals

	Primary stress	Extra stress
ˈ	ˈ	ˈ
ˌ	ˌ	ˌ
ː	ː	ː
ˑ	ˑ	ˑ
˒	˒	˒
˓	˓	˓
˔	˔	˔
˕	˕	˕
˖	˖	˖
˗	˗	˗
˘	˘	˘
˙	˙	˙
˚	˚	˚
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˝	˝	˝
˞	˞	˞
˟	˟	˟
ˠ	ˠ	ˠ
ˡ	ˡ	ˡ
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diacritics

SYLLABICITY & RELEASES	PHONATION	PRIMARY ARTICULATION	SECONDARY ARTICULATION
ɱ̥	Syllabic	ɱ̥	Dental
ɱ̥̊	Non-syllabic	ɱ̥̊	Apical
ɱ̥̊̊	(Pre)aspirated	ɱ̥̊̊	Laminal
ɱ̥̊̊̊	Nasal release	ɱ̥̊̊̊	Advanced
ɱ̥̊̊̊̊	Lateral release	ɱ̥̊̊̊̊	Retracted
ɱ̥̊̊̊̊̊	No audible release	ɱ̥̊̊̊̊̊	Centralized
ɱ̥̊̊̊̊̊̊	Lowered (β̥ is a bilabial approximant)	ɱ̥̊̊̊̊̊̊	Raised (ɹ̥ is a voiced alveolar non-sibilant fricative, ɹ̥̥ a fricative trill)

The authors of textbooks or similar publications often create revised versions of the IPA chart to express their own preferences or needs. The image displays one such version. All pulmonic consonants are moved to the consonant chart. Only the black symbols are on the official IPA chart; additional symbols are in grey. The grey fricatives are part of the extIPA, and the grey retroflex letters are mentioned or implicit in the *Handbook*. The grey click is a retired IPA letter that is still in use.

[ɪntənəʃənəl]

[ɪ̃ɹ̃ənəʃɪnəl]

Phonetic transcriptions of the word *international* in two English dialects

comfortable for readers of English and to be more acceptable across dialects, without the implication of a preferred pronunciation that the IPA might convey. For example, the respelling systems in many American dictionaries (such as *Merriam-Webster*) use ⟨y⟩ for IPA [j] and ⟨sh⟩ for IPA [ʃ], reflecting the usual spelling of those sounds in English.^{[50][51][note 14]} (In IPA, [y] represents the sound of the French ⟨u⟩, as in *tu*, and [sh] represents the sequence of consonants in *grasshopper*.)

Other languages

The IPA is also not universal among dictionaries in languages other than English. Monolingual dictionaries of languages with phonemic orthographies generally do not bother with indicating the pronunciation of most words, and tend to use respelling systems for words with unexpected pronunciations. Dictionaries produced in Israel use the IPA rarely and sometimes use the Hebrew alphabet for transcription of foreign words.^[note 15] Bilingual dictionaries that translate from foreign languages into Russian usually employ the IPA, but monolingual Russian dictionaries occasionally use pronunciation respelling for foreign words.^[note 16] The IPA is more common in bilingual dictionaries, but there are exceptions here too. Mass-market bilingual Czech dictionaries, for instance, tend to use the IPA only for sounds not found in Czech.^[note 17]

Standard orthographies and case variants

IPA letters have been incorporated into the alphabets of various languages, notably via the Africa Alphabet in many sub-Saharan languages such as Hausa, Fula, Akan, Gbe languages, Manding languages, Lingala, etc. Capital case variants have been created for use in these languages. For example, Kabiyè of northern Togo has Ð ɖ, Ŋ ŋ, Ƴ ʏ, ɔ ɔ, ɛ ɛ, ʊ ʊ. These, and others, are supported by Unicode, but appear in Latin ranges other than the IPA extensions.

In the IPA itself, however, only lower-case letters are used. The 1949 edition of the IPA handbook indicated that an asterisk ⟨*⟩ might be prefixed to indicate that a word was a proper name,^[53] and this convention was used by *Le Maître Phonétique*, which was written in IPA rather than in English or French orthography, but it was not included in the 1999 *Handbook*, which notes the contrary use of the asterisk as a placeholder for a sound or feature that does not have a symbol.^[54]

Classical singing

The IPA has widespread use among classical singers during preparation as they are frequently required to sing in a variety of foreign languages. They are also taught by vocal coaches to perfect diction and improve tone quality and tuning.^[55] Opera librettos are authoritatively transcribed in IPA, such as Nico Castel's volumes^[56] and Timothy Cheek's book *Singing in Czech*.^[57] Opera singers' ability to read IPA was used by the site *Visual Thesaurus*, which employed several opera singers "to make recordings for the 150,000 words and phrases in VT's lexical database ... for their vocal stamina, attention to the details of enunciation, and most of all, knowledge of IPA".^[58]

Letters

The International Phonetic Association organizes the letters of the IPA into three categories: pulmonic consonants, non-pulmonic consonants, and vowels.^{[note 18][60][61]}

Pulmonic consonant letters are arranged singly or in pairs of voiceless (tenuis) and voiced sounds, with these then grouped in columns from front (labial) sounds on the left to back (glottal) sounds on the right. In official publications by the IPA, two columns are omitted to save space, with the letters listed among "other symbols" even though theoretically they belong in the main chart.^[note 19] They are arranged in rows from full closure

(occlusives: stops and nasals) at top, to brief closure (vibrants: trills and taps), to partial closure (fricatives), and finally minimal closure (approximants) at bottom, again with a row left out to save space. In the table below, a slightly different arrangement is made: All pulmonic consonants are included in the pulmonic-consonant table, and the vibrants and laterals are separated out so that the rows reflect the common lenition pathway of *stop* → *fricative* → *approximant*, as well as the fact that several letters pull double duty as both fricative and approximant; affricates may then be created by joining stops and fricatives from adjacent cells. Shaded cells represent articulations that are judged to be impossible or not distinctive.

Vowel letters are also grouped in pairs – of unrounded and rounded vowel sounds – with these pairs also arranged from front on the left to back on the right, and from maximal closure at top to minimal closure at bottom. No vowel letters are omitted from the chart, though in the past some of the mid central vowels were listed among the "other symbols".

Consonants

Pulmonic consonants

A pulmonic consonant is a consonant made by obstructing the glottis (the space between the vocal folds) or oral cavity (the mouth) and either simultaneously or subsequently letting out air from the lungs. Pulmonic consonants make up the majority of consonants in the IPA, as well as in human language. All consonants in English fall into this category.^[63]

The pulmonic consonant table, which includes most consonants, is arranged in rows that designate manner of articulation, meaning how the consonant is produced, and columns that designate place of articulation, meaning where in the vocal tract the consonant is produced. The main chart includes only consonants with a single place of articulation.

Place →	Labial			Coronal						Dorsal			Laryngeal	
	Bilabial	Labio-dental	Linguo-labial	Dental	Alveolar	Post-alveolar	Retro-flex	Palatal	Velar	Uvular	Pharyngeal/epiglottal	Glottal		
Manner ↓														
Nasal	<u>m</u> <u>m</u>	<u>ɱ</u> <u>ɱ</u>	<u>ɹ̠</u>	<u>ɲ</u> <u>ɲ</u>	<u>ɳ</u> <u>ɳ</u>	<u>ɳ̠</u> <u>ɳ̠</u>	<u>ɳ̠</u> <u>ɳ̠</u>	<u>ɲ̠</u> <u>ɲ̠</u>	<u>ɲ̠</u> <u>ɲ̠</u>	<u>ɴ</u> <u>ɴ</u>				
Plosive	<u>p</u> <u>b</u>	<u>ɸ</u> <u>β</u>	<u>t̪</u> <u>d̪</u>	<u>t̪</u> <u>d̪</u>	<u>t̪</u> <u>d̪</u>		<u>ʈ</u> <u>ɖ</u>	<u>c</u> <u>ɟ</u>	<u>k</u> <u>g</u>	<u>q</u> <u>ɢ</u>	<u>ʕ</u>	<u>ʔ</u>		
Sibilant fricative				<u>s</u> <u>z</u>	<u>ʃ</u> <u>ʒ</u>	<u>ʃ̠</u> <u>ʒ̠</u>	<u>ʂ</u> <u>ʐ</u>	<u>ɕ</u> <u>ɟ̞</u>						
Non-sibilant fricative	<u>ɸ</u> <u>β</u>	<u>f</u> <u>v</u>	<u>θ</u> <u>ð</u>	<u>θ</u> <u>ð</u>	<u>θ̠</u> <u>ð̠</u>	<u>ɬ</u> <u>ɮ</u>	<u>ɬ̠</u> <u>ɮ̠</u>	<u>ɕ̠</u> <u>ɟ̞̠</u>	<u>x</u> <u>ɣ</u>	<u>χ</u> <u>ʁ</u>	<u>ħ</u> <u>ʕ</u>	<u>h</u> <u>ɦ</u>		
Approximant	<u>ɸ̞</u>	<u>ɸ̞̞</u>		<u>θ̞</u>	<u>ɬ̞</u>	<u>ɬ̞̞</u>	<u>ɬ̞̞̞</u>	<u>ɕ̞</u>	<u>ɰ</u>	<u>ʁ̞</u>		<u>ʔ̞</u>		
Tap/flap	<u>ɸ̞̞̞</u>	<u>ɸ̞̞̞̞</u>	<u>ɸ̞̞</u>											

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Notes

- In rows where some letters appear in pairs (the obstruents), the letter to the right represents a voiced consonant, except breathy-voiced [h].^[64] In the other rows (the sonorants), the single letter represents a voiced consonant.

- While IPA provides a single letter for the coronal places of articulation (for all consonants but fricatives), these do not always have to be used exactly. When dealing with a particular language, the letters may be treated as specifically dental, alveolar, or post-alveolar, as appropriate for that language, without diacritics.
- Shaded areas indicate articulations judged to be impossible.
- The letters [β, ð, ɸ, ɣ, ʁ] are canonically voiced fricatives but may be used for approximants.^[note 20]
- In many languages, such as English, [h] and [ɦ] are not actually glottal, fricatives, or approximants. Rather, they are bare phonation.^[66]
- It is primarily the shape of the tongue rather than its position that distinguishes the fricatives [ʃ ʒ], [ç ʝ], and [ʂ ʐ].
- [ɬ, ʎ] are defined as epiglottal fricatives under the "Other symbols" section in the official IPA chart, but they may be treated as trills at the same place of articulation as [h, ɦ] because trilling of the aryepiglottic folds typically co-occurs.^[67]
- Some listed phones are not known to exist as phonemes in any language.

Non-pulmonic consonants

Non-pulmonic consonants are sounds whose airflow is not dependent on the lungs. These include clicks (found in the Khoisan languages and some neighboring Bantu languages of Africa), implosives (found in languages such as Sindhi, Hausa, Swahili and Vietnamese), and ejectives (found in many Amerindian and Caucasian languages).

		<u>BL</u> bilabial	<u>LD</u> labiodental	<u>D</u> dental	<u>A</u> alveolar	<u>PA</u> postalveolar	<u>RF</u> retroflex	<u>P</u> labial	<u>V</u> labiodental	<u>U</u> uvular
<u>Ejective</u>	<u>Stop</u>	p′			t′		t′	c′	k′	q′
	<u>Fricative</u>	ɸ′	f′	θ′	s′	ʃ′	ʂ′	ɸ′	x′	χ′
	<u>Lateral fricative</u>				ɬ′					
<u>Click</u> (top: velar; bottom: uvular)	<u>Tenuis</u>	k̠ q̠		k̠ q̠	k̠ q̠		k̠ q̠	k̠ q̠		
	<u>Voiced</u>	g̠ ɠ̠		g̠ ɠ̠	g̠ ɠ̠		g̠ ɠ̠	g̠ ɠ̠		
	<u>Nasal</u>	ŋ̠ ɴ̠		ŋ̠ ɴ̠	ŋ̠ ɴ̠		ŋ̠ ɴ̠	ŋ̠ ɴ̠		
	<u>Tenuis lateral</u>				k̠l q̠l					
	<u>Voiced lateral</u>				g̠l ɠ̠l					
	<u>Nasal lateral</u>				ŋ̠l ɴ̠l					
<u>Implosive</u>	<u>Voiced</u>	ɓ			ɗ		ɗ̠	ɟ	ɠ	ɠ̠
	<u>Voiceless</u>	ɓ̥			ɗ̥		ɗ̠̥	ɟ̥	ɠ̥	ɠ̠̥

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Notes

- Clicks have traditionally been described as consisting of a forward place of articulation, commonly called the click "type" or historically the "influx", and a rear place of articulation, which when combined with the quality of the click is commonly called the click "accompaniment" or historically the "efflux". The IPA click letters indicate only the click type (forward articulation and release). Therefore, all clicks require two letters for proper notation: ⟨k̠, ɡ̠, q̠⟩, etc., or with the order reversed if both the forward and rear releases are audible. The letter for the rear articulation is frequently omitted, in which case a ⟨k⟩ may usually be assumed. However, some researchers dispute the idea that clicks should be analyzed as doubly

articulated, as the traditional transcription implies, and analyze the rear occlusion as solely a part of the airstream mechanism.^[68] In transcriptions of such approaches, the click letter represents both places of articulation, with the different letters representing the different click types, and diacritics are used for the elements of the accompaniment: ⟨l̥, ɺ, ɿ̥⟩, etc.

- Letters for the voiceless implosives ⟨ɓ, ɗ, ɠ, ʄ, ɛ̃⟩ are no longer supported by the IPA, though they remain in Unicode. Instead, the IPA typically uses the voiced equivalent with a voiceless diacritic: ⟨ɓ̥, ɗ̥⟩, etc.
- The letter for the retroflex implosive, ⟨ɗ̠⟩, is not "explicitly IPA approved",^[69] but the IPA has endorsed the inclusion of ⟨ɗ̠⟩ and voiceless ⟨ɗ̠̥⟩ into Unicode.
- The ejective diacritic is placed at the right-hand margin of the consonant, rather than immediately after the letter for the stop: ⟨t͡ʃʼ⟩, ⟨kʷʼ⟩. In imprecise transcription, it often stands in for a superscript glottal stop in glottalized but pulmonic sonorants, such as [mʼ], [lʼ], [wʼ], [aʼ] – also transcribable as creaky [ṁ], [Ḻ], [Ṃ], [Ḑ].

Affricates

Affricates and co-articulated stops are represented by two letters in sequence. For clarity, this digraph may be joined by a tie bar, which may appear either above or below the letters with no difference in meaning.^[note 21] Affricates are optionally represented by ligatures – e.g. ⟨t͡ʃ, d͡ʒ⟩ – though this is no longer official IPA usage.^[1] Alternatively, a superscript notation for a consonant release is sometimes used to transcribe affricates, for example ⟨t͡ʃ̚⟩ for [t͡ʃ̚], although in precise notation this would indicate a fricative release rather than an affricate. The letters for the palatal plosives ⟨c⟩ and ⟨ɟ⟩ are often used as a convenience for [t͡ɕ] and [d͡ʝ] or similar affricates, even in official IPA publications, so they must be interpreted with care.^[71]

		Bilabial	Labio-dental	Dental	Alveolar	Post-alveolar	Retroflex	Palatal	Velar	Uvular	Epi-glottal	Glottal
Pulmonic	Sibilant				ts dz	t͡ʃ d͡ʒ	t͡ɕ d͡ʝ	t͡ɕ d͡ʝ				
	Non-sibilant	pɸ bβ	pɸ bβ	t̪ θ d̪ ð	t̪ t̪ d̪ d̪	t̪ t̪ d̪ d̪		cç ɟʝ	kx gɣ	qχ ɢʁ	ʔḥ ʔʕ	ʔh
	Lateral				tɬ dɮ		tɬ dɮ	cɕ ɟʝ	kɕ gɕ			
Ejective	Central			t̪ʰ	tsʰ	t͡ʃʰ	t͡ɕʰ		kxʰ	qχʰ		
	Lateral				tɬʰ			cɕʰ	kɕʰ			

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Because in a true affricate the plosive element and the fricative element are homorganic, and the place of articulation of an affricate is most audible in the fricative element, the letter for the former will not always be precisely transcribed where such precision would be redundant. For example, while the English *ch* sound is [t͡ʃ̚] in close transcription, the diacritic is commonly left off, for [t͡ʃ̚]. Similarly, [t͡ʃ̚] and [d͡ʒ̚] are more commonly written [t͡ʃ̚] and [d͡ʒ̚], and in the ligatures there is only a single retroflex hook.

Co-articulated consonants

Co-articulated consonants are sounds that involve two simultaneous places of articulation (are pronounced using two parts of the vocal tract). In English, the [w] in "went" is a coarticulated consonant, being pronounced by rounding the lips and raising the back of the tongue. Similar sounds are [ɰ] and [ɥ]. In some languages, plosives can be double-articulated, for example in the name of Laurent Gbagbo.

Nasal	Fricative/approximant
$\widehat{\text{nm}}$ Labial–alveolar	$\underset{\cdot}{\text{ɸ}}$ $\underset{\cdot}{\text{ɸ}}$ Labialized palatal
$\widehat{\eta\text{m}}$ Labial–retroflex	$\underset{\cdot}{\text{m}}$ $\underset{\cdot}{\text{w}}$ Labialized velar
$\widehat{\eta\text{m}}$ Labial–velar	$\underset{\cdot}{\text{h}}$ <i>Sj</i> -sound (variable)
Plosive	Lateral approximant
$\widehat{\text{tp}}$ $\widehat{\text{db}}$ Labial–alveolar	$\underset{\cdot}{\text{ɭ}}$ Velarized alveolar
$\widehat{\text{t̠p}}$ $\widehat{\text{q̠b}}$ Labial–retroflex	Implosive
$\widehat{\text{kp}}$ $\widehat{\text{gb}}$ Labial–velar	$\underset{\cdot}{\text{ɓ}}$ $\underset{\cdot}{\text{ɓ}}$ Labial–velar
$\widehat{\text{qʔ}}$ Uvular–epiglottal	Ejective
$\widehat{\text{qp}}$ Labial–uvular	$\widehat{\text{tp}}'$ Labial–alveolar

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Notes

- $[\text{h}]$, the *Swedish sj*-sound, is described by the IPA as a "simultaneous $[\text{ʃ}]$ and $[\text{x}]$ ", but it is unlikely such a simultaneous fricative actually exists in any language.^[72]
- Multiple tie bars can be used: $\langle \widehat{\text{abc}} \rangle$ or $\langle \text{abc} \rangle$. For instance, a pre-voiced velar affricate may be transcribed as $\langle \widehat{\text{gkx}} \rangle$
- If a diacritic needs to be placed on or under a tie bar, the *combining grapheme joiner* (U+034F) needs to be used, as in $[\text{bd̪əbd̪}]$ 'chewed' (Margi). Font support is spotty, however.

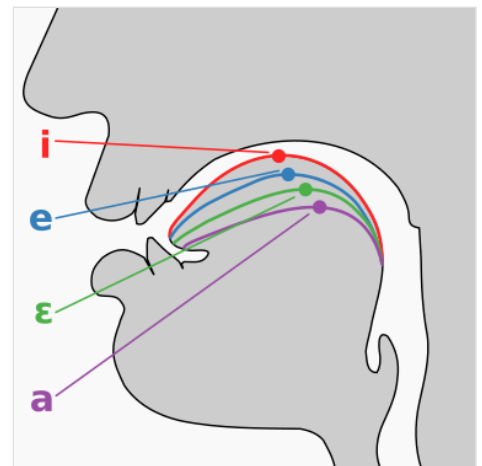
With the implosives, authors may not bother to redundantly mark both letters as implosive, but instead write them as less-cluttered $\langle \widehat{\text{gb}} \rangle$ and even $\langle \text{k} \square \rangle$.

Vowels

The IPA defines a vowel as a sound which occurs at a syllable center.^[73] Below is a chart depicting the vowels of the IPA. The IPA maps the vowels according to the position of the tongue.

	Front	Central	Back
Close	$\text{i} \cdot \text{y}$	$\text{ɨ} \cdot \text{ɯ}$	$\text{ɯ} \cdot \text{u}$
Near-close	$\text{ɪ} \cdot \text{ʏ}$		ʊ
Close-mid	$\text{e} \cdot \text{ø}$	$\text{ə} \cdot \text{ɘ}$	$\text{ɤ} \cdot \text{o}$
Mid	$\text{ɛ} \cdot \text{ø}$	ə	$\text{ɤ} \cdot \text{o}$
Open-mid	$\text{ɛ} \cdot \text{œ}$	$\text{ɜ} \cdot \text{ɞ}$	$\text{ʌ} \cdot \text{ɔ}$
Near-open	æ	ɐ	
Open	$\text{a} \cdot \text{œ} - \text{ä} \cdot$		$\text{ɑ} \cdot \text{ɒ}$

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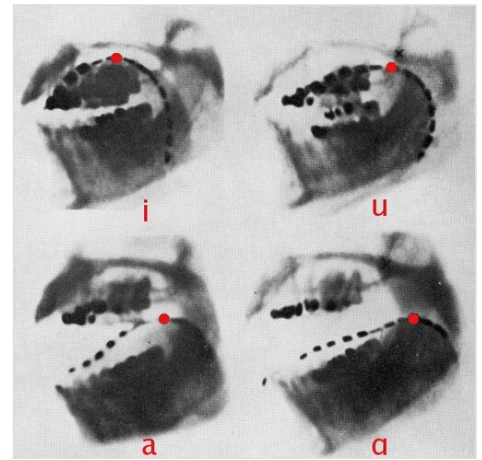


Tongue positions of cardinal front vowels, with highest point indicated. The position of the highest point is used to determine vowel height and backness.

The vertical axis of the chart is mapped by vowel height. Vowels pronounced with the tongue lowered are at the bottom, and vowels pronounced with the tongue raised are at the top. For example, [a] (the first vowel in *father*) is at the bottom because the tongue is lowered in this position. [i] (the vowel in "meet") is at the top because the sound is said with the tongue raised to the roof of the mouth.

In a similar fashion, the horizontal axis of the chart is determined by vowel backness. Vowels with the tongue moved towards the front of the mouth (such as [ɛ], the vowel in "met") are to the left in the chart, while those in which it is moved to the back (such as [ʌ], the vowel in "but") are placed to the right in the chart.

In places where vowels are paired, the right represents a rounded vowel (in which the lips are rounded) while the left is its unrounded counterpart.



X-ray photos show the sounds [i, u, a, ɑ].

Diphthongs

Diphthongs may be written as simple sequences of letters, but for clarity they are commonly specified with a non-syllabic diacritic, as in ⟨ui̯⟩ or ⟨ui̯̯⟩, or with a superscript for the on- or off-glide, as in ⟨u^ɪ⟩ or ⟨^ui̯⟩. Sometimes a tie bar is used: ⟨ui̯̯̯⟩, especially when it is difficult to tell if the diphthong is characterized by an on-glide or an off-glide, or when it is variable.

Notes

- ⟨a⟩ officially represents a front vowel, but there is little if any distinction between front and central open vowels (see Vowel § Acoustics), and ⟨a⟩ is frequently used for an open central vowel.^[48] If disambiguation is required, the retraction diacritic or the centralized diacritic may be added to indicate an open central vowel, as in ⟨a̠⟩ or ⟨ä̯⟩.

Diacritics and prosodic notation

Diacritics are used for phonetic detail. They are added to IPA letters to indicate a modification or specification of that letter's normal pronunciation.^[74]

By being made superscript, any IPA letter may function as a diacritic, conferring elements of its articulation to the base letter.^[75] Those superscript letters listed below are specifically provided for by the IPA *Handbook*; other uses can be illustrated with ⟨t̚⟩ ([t] with fricative release), ⟨s̟⟩ ([s] with affricate onset), ⟨ⁿd̚⟩ (prenasalized [d]), ⟨b̤⟩ ([b] with breathy voice), ⟨m̤⟩ (glottalized [m]), ⟨s̺⟩ ([s] with a flavor of [ʃ], i.e. a voiceless alveolar retracted sibilant), ⟨o̯̥⟩ ([o] with diphthongization), ⟨u̯̥⟩ (compressed [u]). Superscript diacritics placed after a letter are ambiguous between simultaneous modification of the sound and phonetic detail at the end of the sound. For example, labialized ⟨k^w⟩ may mean either simultaneous [k] and [w] or else [k] with a labialized release. Superscript diacritics placed before a letter, on the other hand, normally indicate a modification of the onset of the sound (⟨m̤⟩ glottalized [m], ⟨^ʔm̤⟩ [m] with a glottal onset). (See § Superscript IPA.)

Airstream diacritics					
	k' s'	<u>Ejective</u>			
Syllabicity diacritics					
	ɹ̥ ɲ̥	<u>Syllabic</u>		ɿ ʊ̯	<u>Non-syllabic</u>
	ɹ̥̄ ɲ̥̄			ɿ̄ ʊ̯̄	
Consonant-release diacritics					
	tʰ	<u>Aspirated</u> ^[a]		p̚	<u>No audible release</u>
	dⁿ	<u>Nasal release</u>		dˡ	<u>Lateral release</u>
	tʰ	Voiceless dental fricative release		tˣ	Voiceless velar fricative release
	dᵑ	Mid central vowel release			
Phonation diacritics					
	ɳ̥ ɖ̥	<u>Voiceless</u>		ʂ̚ ʈ̚	<u>Voiced</u>
	ɳ̥̄ ɖ̥̄				
	b̤ a̤	<u>Breathy voiced</u> ^[a]		b̰ a̰	<u>Creaky voiced</u>
Articulation diacritics					
	t̪ d̪	Dental (◌̪ is <u>dentolabial</u> or underbite in extIPA)		t̠ d̠	<u>Linguolabial</u>
	t̪̄ d̪̄				
	t̺ d̺	<u>Apical</u>		t̻ d̻	<u>Laminal</u>
	ɥ̟ ʈ̟	<u>Advanced (fronted)</u>		ɪ̠ ʈ̠	<u>Retracted (backed)</u>

				q ^ɤ _[β]	
	ẽ ä	<u>Centralized</u>		ě ů	<u>Mid-centralized</u>
	e _ɪ r _ɪ	Raised ([ɾ], [ɭ] are fricatives)		e _ɪ β _ɪ	Lowered ([β], [ɣ _ɪ] are approximants)
	l _ɪ			y _ɪ ɣ _ɪ	
Co-articulation diacritics					
	ɔ̹ ɣ̹	More rounded / less spread (over-rounding)		ɔ̹ x ^w _ɛ	Less rounded / more spread (under-rounding) ^[ɹ]
	ý Ẃ			ý Ẃ ^w	
	t ^w d ^w	<u>Labialized</u>		tʲ dʲ	<u>Palatalized</u>
	tʲ dʲ	<u>Velarized</u>		ɫ z	Velarized or pharyngealized
	tʳ aʳ	<u>Pharyngealized</u>			
	e _ɪ o _ɪ	Advanced tongue root (expanded pharynx)		e _ɪ o _ɪ	<u>Retracted tongue root</u>
	y□			y□	
	ẽ ẓ	<u>Nasalized</u>		ə̃ ʒ̃	<u>Rhoticity</u>

Notes:

- α. With aspirated voiced consonants, the aspiration is usually also voiced (voiced aspirated – but see voiced consonants with voiceless aspiration). Many linguists prefer one of the diacritics dedicated to breathy voice over simple aspiration, such as <ɸ̤>. Some linguists restrict that diacritic to sonorants, such as breathy-voice <ṁ>, and transcribe voiced-aspirated obstruents as e.g. <bʰ>.
- β. In the Unicode Pipeline for 2025. Care must be taken that a superscript retraction sign is not mistaken for a macron for mid tone.
- γ. These are relative to the cardinal value of the letter. They can also apply to unrounded vowels: [ɛ̹] is more spread (less rounded) than cardinal [ɛ], and [u̹] is less spread than cardinal [u].^[76]
Since <x^w> can mean that the [x] is labialized (rounded) throughout its articulation, and <ɣ̹> makes no sense ([x] is already completely unrounded), <x^w> can only mean a less-labialized/rounded [x^w]. However, readers might mistake <ɣ̹^w> for "[x̣]" with a labialized off-glide, or might wonder if the two diacritics cancel each other out. Placing the 'less rounded' diacritic under the labialization diacritic, <x^w_ɛ>, makes it clear that it is the labialization that is 'less rounded' than its cardinal IPA value.

A diacritic may be moved to avoid conflicts of space. One that is normally placed below a letter may be moved above it to avoid a descender or another diacritic, as with the voiceless ring on ⟨ŋ̥⟩, and vice versa with the tie bar on ⟨ts̺⟩, though the tie bar is basically in free variation.^[74] Exceptions are the tilde, trema and caron/wedge – and, in extIPA, the bridge – which are defined differently when placed above and below a letter. The raising and lowering diacritics have a third option of being set as spacing forms ⟨˘˙⟩, ⟨˙˘⟩ that avoid both descenders and ascenders.

A couple additional superscript letters are found for secondary articulation. In the *Handbook*, for example, ⟨^h⟩ is used for voiced aspiration. ⟨^q⟩ is commonly seen with languages such as Twi where consonants may be simultaneously palatalized and labialized, while ⟨^ʔ⟩ may be used for glottalized sounds without specifying whether they are ejective or have creaky voice. ExtIPA provides ⟨^{*}⟩ for uvularization, and the Voice Quality Symbols provide a couple more. However, only limited set of IPA letters are used in this fashion; for all others, superscripting indicates more ambiguous shading of the sound.

The state of the glottis can be finely transcribed with diacritics. A series of alveolar plosives ranging from open-glottis to closed-glottis phonation is:

Phonation scale		
Open glottis	[t̟]	<u>voiceless</u>
	[d̟]	<u>breathy voice</u> , also called <i>murmured</i>
	[d̠]	<u>slack voice</u>
Sweet spot	[d̡]	<u>modal voice</u>
	[d̢]	<u>stiff voice</u>
	[ḍ]	<u>creaky voice</u>
Closed glottis	[ʔ̚]	glottal closure

Additional diacritics are provided by the Extensions to the IPA for speech pathology.

Suprasegmentals

These symbols describe the features of a language above the level of individual consonants and vowels, that is, at the level of syllable, word or phrase. These include prosody, pitch, length, stress, intensity, tone and gemination of the sounds of a language, as well as the rhythm and intonation of speech.^[77] Various ligatures of pitch/tone letters and diacritics are provided for by the Kiel Convention and used in the IPA *Handbook* despite not being found in the summary of the IPA alphabet found on the one-page chart.

Under capital letters below we will see how a carrier letter may be used to indicate suprasegmental features such as labialization or nasalization. Some authors omit the carrier letter, for e.g. suffixed [k^hu^{x̺}ts̺]^w or prefixed [w^kh^ux̺ts̺],^[note 22] or place a spacing variant of a diacritic such as ⟨˘˙⟩ or ⟨˙˘⟩ at the beginning or end of a word to indicate that it applies to the entire word.^[note 23]

Length, stress, and rhythm			
'ke	Primary stress (appears before stressed syllable)	ˌke	Secondary stress (appears before stressed syllable)
eː kː	Long (long vowel or geminate consonant)	eˑ	Half-long
		ě ċ	Extra-short; flap
ek.ste eks.te	Syllable break (internal boundary)	es_e	Linking (lack of a boundary; a phonological word) ^[note 24]
Intonation			
↓[α]	Minor or foot break	↓↓[α]	Major or intonation break
↗	Global rise ^[note 25]	↘	Global fall ^[note 25]
Up- and down-step			
↑ke	Upstep	↓ke	Downstep

Notes:

- α. The pipes for intonation breaks should be a heavier weight than the letters for click consonants. Because fonts do not reflect this, the intonation breaks in the official IPA charts are set in bold typeface.

Pitch diacritics ^[note 26]					
ᵿ ẽ	Extra high	ᵿ ẽ	Rising	ᵿ ẽ	Mid-rising
ᵿ é	High	ᵿ ê	Falling	ᵿ ẽ	Low-rising
ᵿ ē	Mid	ᵿ ẽ	Peaking (rising–falling) ^[note 27]	ᵿ ê	High-falling
ᵿ è	Low	ᵿ ẽ	Dipping (falling–rising) ^[note 27]	ᵿ ẽ	Mid-falling
ᵿ ẽ	Extra low	(etc.) ^[note 28]			

Chao tone letters ^[note 26]				
ᵿe	ᵿe	eᵿ	eᵿ	High
ᵿe	ᵿe	eᵿ	eᵿ	Half-high
ᵿe	ᵿe	eᵿ	eᵿ	Mid
ᵿe	ᵿe	eᵿ	eᵿ	Half-low
ᵿe	ᵿe	eᵿ	eᵿ	Low
ᵿe	ᵿe	eᵿ	eᵿ	Rising (low to high or generic)
ᵿe	ᵿe	eᵿ	eᵿ	Falling (high to low or generic)
(etc.)				

The old staveless tone letters, which are effectively obsolete, include high <ˊe>, mid <–e> [not supported by Unicode], low <_e>, rising <ˑe>, falling <ˋe>, low rising <ˊ_e> and low falling <ˋ_e>.

Stress

Officially, the stress marks ⟨'⟩, ⟨ˈ⟩ appear before the stressed syllable, and thus mark the syllable boundary as well as stress (though the syllable boundary may still be explicitly marked with a period).^[80] Occasionally the stress mark is placed immediately before the nucleus of the syllable, after any consonantal onset.^[81] In such transcriptions, the stress mark does not mark a syllable boundary. The primary stress mark may be doubled ⟨''⟩ for extra stress (such as prosodic stress). The secondary stress mark is sometimes seen doubled ⟨ˈˈ⟩ for extra-weak stress, but this convention has not been adopted by the IPA.^[80] Some dictionaries place both stress marks before a syllable, ⟨!⟩, to indicate that pronunciations with either primary or secondary stress are heard, though this is not IPA usage.^[note 29]

Boundary markers

There are three boundary markers: ⟨.⟩ for a syllable break, ⟨|⟩ for a minor prosodic break and ⟨||⟩ for a major prosodic break. The tags 'minor' and 'major' are intentionally ambiguous. Depending on need, 'minor' may vary from a foot break to a break in list-intonation to a continuing–prosodic unit boundary (equivalent to a comma), and while 'major' is often any intonation break, it may be restricted to a final–prosodic unit boundary (equivalent to a period). The 'major' symbol may also be doubled, ⟨|||⟩, for a stronger break.^[note 30]

Although not part of the IPA, the following additional boundary markers are often used in conjunction with the IPA: ⟨μ⟩ for a mora or mora boundary, ⟨σ⟩ for a syllable or syllable boundary, ⟨+⟩ for a morpheme boundary, ⟨#⟩ for a word boundary (may be doubled, ⟨##⟩, for e.g. a breath-group boundary),^[83] ⟨\$⟩ for a phrase or intermediate boundary and ⟨%⟩ for a prosodic boundary. For example, C# is a word-final consonant, %V a post-pausa vowel, and σC a syllable-initial consonant.

Pitch and tone

⟨↑ ↓⟩ are defined in the *Handbook* as "upstep" and "downstep", concepts from tonal languages. However, the upstep symbol can also be used for pitch reset, and the IPA *Handbook* uses it for prosody in the illustration for Portuguese, a non-tonal language.

Phonetic pitch and phonemic tone may be indicated by either diacritics placed over the nucleus of the syllable – e.g., high-pitch ⟨é⟩ – or by Chao tone letters placed either before or after the word or syllable. There are three graphic variants of the tone letters: with or without a stave, and facing left or facing right from the stave. The stave was introduced with the 1989 Kiel Convention, as was the option of placing a staved letter after the word or syllable, while retaining the older conventions. There are therefore six ways to transcribe pitch/tone in the IPA: i.e., ⟨é⟩, ⟨le⟩, ⟨el⟩, ⟨fe⟩, ⟨ef⟩ and ⟨¯e⟩ for a high pitch/tone.^{[80][84][85]} Of the tone letters, only left-facing staved letters and a few representative combinations are shown in the summary on the *Chart*, and in practice it is currently more common for tone letters to occur after the syllable/word than before, as in the Chao tradition. Placement before the word is a carry-over from the pre-Kiel IPA convention, as is still the case for the stress and upstep/downstep marks. The IPA endorses the Chao tradition of using the left-facing tone letters, ⟨l 1 1 1 J⟩, for underlying tone, and the right-facing letters, ⟨f ɿ ɿ ɿ L⟩, for surface tone, as occurs in tone sandhi, and for the intonation of non-tonal languages. In the Portuguese illustration in the 1999 *Handbook*, for example, tone letters are placed before a word or syllable to indicate prosodic pitch (equivalent to [↗] global rise and [↘] global fall, but allowing more precision), and in the Cantonese illustration they are placed after a word/syllable to indicate lexical tone. Theoretically therefore prosodic pitch and lexical tone could be simultaneously transcribed in a single text, though this is not a formalized distinction.

Rising and falling pitch, as in contour tones, are indicated by combining the pitch diacritics and letters in the table, such as grave plus acute for rising [ě] and acute plus grave for falling [ê]. Only six combinations of two diacritics are supported, and only across three levels (high, mid, low), despite the diacritics supporting five

The Chao tone letters, on the other hand, may be combined in any pattern, and are therefore used for more complex contours and finer distinctions than the diacritics allow, such as mid-rising [eʔ], extra-high falling [e̞ʔ], etc. There are 20 such possibilities. However, in Chao's original proposal, which was adopted by the IPA in 1989, he stipulated that the half-high and half-low letters ⟨1 4⟩ may be combined with each other, but not with the other three tone letters, so as not to create spuriously precise distinctions. With this restriction, there are 8 possibilities.^[86]

Although tone diacritics and tone letters are presented as equivalent on the chart, "this was done only to simplify the layout of the chart. The two sets of symbols are not comparable in this way."^[87] Using diacritics, a high tone is ⟨é⟩ and a low tone is ⟨è⟩; in tone letters, these are ⟨eḽ⟩ and ⟨eḷ⟩. One can double the diacritics for extra-high ⟨ě⟩ and extra-low ⟨ě̀⟩; there is no parallel to this using tone letters. Instead, tone letters have mid-high ⟨eḽ⟩ and mid-low ⟨eḷ⟩; again, there is no equivalent among the diacritics. Thus in a three-register tone system, ⟨é ē è⟩ are equivalent to ⟨eḽ eḷ eḷ⟩, while in a four-register system, ⟨ě é è è̀⟩ may be equivalent to ⟨eḽ eḷ eḷ eḷ⟩.^[80]

The correspondence breaks down even further once they start combining. For more complex tones, one may combine three or four tone diacritics in any permutation,^[80] though in practice only generic peaking (rising-falling) $\overset{\curvearrowright}{\text{e}}$ and dipping (falling-rising) $\underset{\curvearrowleft}{\text{e}}$ combinations are used. Chao tone letters are required for finer detail (e^{V} , e^{L} , e^{VL} , e^{LV} , etc.). Although only 10 peaking and dipping tones were proposed in Chao's original, limited set of tone letters, phoneticians often make finer distinctions, and indeed an example is found on the IPA Chart.^[note 33] The system allows the transcription of 112 peaking and dipping pitch contours, including tones that are level for part of their length.

Original (restricted) set of Chao tone letters^[note 34]

Register	Level [note 35]	Rising	Falling	Peaking	Dipping
eJ	eJJ	eJH	eJL	eJHH	eJLL
eI	eII	eIH	eIL	eIHH	eILL
eU	eUU	eUH	eUL	eUHH	eULL
eT	eTT			eTHL	eTLH
eV	eVV	eVJ	eVL	eVJH	eVJL

Chao tone letters generally appear after each syllable, for a language with syllable tone (e.g. ⟨aᵛᵛᵛ⟩) or after the phonological word, for a language with word tone (e.g. ⟨avᵛᵛ⟩). The IPA gives the option of placing the tone letters before the word or syllable (⟨ᵛᵛᵛa⟩, ⟨ᵛᵛav⟩), and illustrates this for prosody, but it is rare for lexical tone.

Comparative degree

21/41

modifies, e.g. ⟨̃⟩, this can be an aid to legibility, just as it is with the composite superscript c-cedilla ⟨ç⟩ and rhotic vowels ⟨Ṙ⟩. Superscript length marks can be used to indicate the length of aspiration of a consonant, e.g. [p^h t^h k^h]. Another option is to use extIPA parentheses and a doubled diacritic: ⟨p^(h) t^h k^{hh}⟩.^[46]

Obsolete and nonstandard symbols

A number of IPA letters and diacritics have been retired or replaced over the years. This number includes duplicate symbols, symbols that were replaced due to user preference, and unitary symbols that were rendered with diacritics or digraphs to reduce the inventory of the IPA. The rejected symbols are now considered obsolete, though some are still seen in the literature.

The IPA once had several pairs of duplicate symbols from alternative proposals, but eventually settled on one or the other. An example is the vowel letter ⟨ɹ⟩, rejected in favor of ⟨ɜ⟩. Affricates were once transcribed with ligatures, such as ⟨ʧ ʤ⟩ (and others, some of which are not found in Unicode). These have been officially retired but are still used. Letters for specific combinations of primary and secondary articulation have also been mostly retired, with the idea that such features should be indicated with tie bars or diacritics: ⟨ɕ⟩ for [zʷ] is one. In addition, the rare voiceless implosives, ⟨ɓ ɗ ɠ ɡ⟩, were dropped soon after their introduction and are now usually written ⟨ɓ̥ ɗ̥ ɠ̥ ɡ̥⟩. The original set of click letters, ⟨ɀ, Ɂ, ɂ, Ƀ⟩, was retired but is still sometimes seen, as the current pipe letters ⟨ǀ, ǁ, ǂ, ǃ⟩ can cause problems with legibility, especially when used with brackets ([] or /), the letter ⟨ǁ⟩ (small L), or the prosodic marks ⟨ǀ, ǁ⟩. (For this reason, some publications which use the current IPA pipe letters disallow IPA brackets.)^[107]

Individual non-IPA letters may find their way into publications that otherwise use the standard IPA. This is especially common with:

- Affricates, such as the Americanist barred lambda ⟨λ̤⟩ for [tʃ] or ⟨č⟩ for [tʃ].^[note 38]
- The Karlgren letters for Chinese vowels, ⟨ɿ, ʮ, ɥ, ʮ⟩.
- Digits or combinations of digits and letters for tonal phonemes that have conventional numbers in a local tradition, such as the four tones of Standard Chinese. This may be more convenient for comparison between related languages and dialects than a phonetic transcription would be, because tones vary more unpredictably than segmental phonemes do.
- Digits for tone levels, which are simpler to typeset, though the lack of standardization can cause confusion – e.g. ⟨1⟩ is high tone in some languages, but low tone in others; and ⟨3⟩ may be high, medium, or low tone, depending on the local convention.
- Iconic extensions of standard IPA letters that are implicit in the alphabet, such as retroflex ⟨ɖ⟩ and ⟨ɗ⟩. These are referred to in the *Handbook* and have been included in Unicode at IPA request.
- Even presidents of the IPA have used para-IPA notation, such as resurrecting the old diacritic ⟨ɸ⟩ for purely labialized sounds (not simultaneously velarized), the lateral fricative letter ⟨ɬ⟩, and either the old dot diacritic ⟨ʂ ʐ⟩ or the novel letters ⟨ʃ ʒ⟩ for the not-quite-retroflex fricatives of Polish *sz*, *ż* and of Russian *ш*, *ж*.

In addition, it is common to see *ad hoc* typewriter substitutions, generally capital letters, for when IPA support is not available, e.g. S for ⟨ʃ⟩. (See also SAMPA and X-SAMPA substitute notation.)

Extensions

The Extensions to the International Phonetic Alphabet for Disordered Speech, commonly abbreviated "extIPA" and sometimes called "Extended IPA", are symbols whose original purpose was to accurately transcribe disordered speech. At the Kiel Convention in 1989, a group of linguists drew up the initial extensions,^[note 39] which were based on the previous work of the PRDS (Phonetic Representation of Disordered Speech) Group in

the early 1980s.^[109] The extensions were first published in 1990, then modified, and published again in 1994 in the *Journal of the International Phonetic Association*, when they were officially adopted by the ICPLA.^[110] While the original purpose was to transcribe disordered speech, linguists have used the extensions to designate a number of sounds within standard communication, such as hushing, gnashing teeth, and smacking lips,^[2] as well as regular lexical sounds such as lateral fricatives that do not have standard IPA symbols.

In addition to the Extensions to the IPA for disordered speech, there are the conventions of the Voice Quality Symbols, which include a number of symbols for additional airstream mechanisms and secondary articulations in what they call "voice quality".

Associated notation

Capital letters and various characters on the number row of the keyboard are commonly used to extend the alphabet in various ways.

Associated symbols

There are various punctuation-like conventions for linguistic transcription that are commonly used together with IPA. Some of the more common are:

 $\langle * \rangle$

- (a) A reconstructed form.
(b) An ungrammatical form (including an unphonemic form).

< ** >

- (a) A reconstructed form, deeper (more ancient) than a single ⟨*⟩, used when reconstructing even further back from already-starred forms.
- (b) An ungrammatical form. A less common convention than ⟨*⟩ (b), this is sometimes used when reconstructed and ungrammatical forms occur in the same text.^[111]

 $\langle x \rangle, \langle X \rangle$

An ungrammatical form. A less common convention than <*> (b), this is sometimes used when reconstructed and ungrammatical forms occur in the same text.^[112]

<?>

A doubtfully grammatical form.

 $\langle \% \rangle$

A generalized form, such as a typical shape of a wanderwort that has not actually been reconstructed.^[113]

<#>

A word boundary – e.g. $\langle \#V \rangle$ for a word-initial vowel.

<\$>

A phonological word boundary; e.g. ⟨H\$⟩ for a high tone that occurs in such a position.

CONSONANTS (other than those on the IPA Chart)												© 2009 ICPLA 2025	
	Labial	Labio-dental	Labio-alveolar	Dento-labial	Dental	Linguo-labial	Inter-dental	Alveolar	Retroflex	Palatal	Velar	Velopharyngeal	Upper pharyngeal
Plosive		p̥ b̥	p̥ b̥	p̥ b̥	t̥ d̥	t̥ d̥	f̥ ɸ̥						q̥ ɓ̥
Nasal		m̥ n̥	m̥ n̥	m̥ n̥	ɳ̥ ɳ̥	ɳ̥ ɳ̥							
Trill					r̥	r̥						ʀ̥ ʀ̥	
Fricative, median		f̥ y̥	f̥ y̥	θ̥ ð̥	θ̥ ð̥	θ̥ ð̥	θ̥ ð̥	θ̥ ð̥				ʃ̥ ʒ̥	
Fricative, lateral					ɬ̥ ɮ̥	ɬ̥ ɮ̥			ɬ̥ ɮ̥	ɬ̥ ɮ̥	ɬ̥ ɮ̥		
Fricative, lat.-med.								ʂ̥ ʐ̥					
Fricative, nasal	m̥ n̥	m̥ n̥	m̥ n̥					ɳ̥ ɳ̥	ɳ̥ ɳ̥	ɳ̥ ɳ̥	ɳ̥ ɳ̥		
Approximant, lateral						l̥	l̥						
Percussive	ʘ̥				ɶ̥								

DIACRITICS								
ʁ	labial spreading	ɟ ɥ	ɔ̥	partially dental	ɦ	ɰ	main gesture offset right	ɤ
ɣ	strong articulation	ʃ	ɸ	fricative nasal escape	ʋ	ɰ	main gesture offset left	ɤ̃
ɣ̃	weak articulation	ʋ	ɸ̃	velopharyngeal friction	ɤ̃	ɰ̃	whistled articulation	ɤ̃̃
ʁ̃	reiteration	p̃/p̃̃	l̃	ingressive airflow	pl̃	ɰ̃̃	sliding articulation	ɔ̃̃̃

[illegible]

OTHER SOUNDS			
ɹ	apical r	ɕ ɟ ɳ	velodorsal plosives and nasal
ɹ̠	blunched r (molar r)	ɿ	sublaminal lower-alveolar percussive
ʒ ʒ̥	laminal fricatives (incl. lowered tongue tip)	ɿ̥	alveolar click with sublaminal percussive release
ɖ̥ ɣ̥	[ɖ, ɣ] with lateral fricated release	ʋ̥	buccal interdental trill (raspberry)
ɿ̥ ɖ̥	[ɿ, ɖ] with lateral and median release	ʋ̥̥	[ɿ] with bilabial aspiration
ɿ̥̥ ɖ̥̥	linguolabial affricates		• sound with no available symbol

Chart of the Extensions to the International Phonetic Alphabet (extIPA), as of 2025

⟨+⟩

A morpheme boundary; e.g. */'nɛl+t/* for English *knelt*.

⟨_⟩

The location of a segment – e.g. ⟨V_ V⟩ for an intervocalic position, or ⟨_#⟩ for word-final position.

⟨~⟩

Alternation or contrast – e.g. [f] ~ [v] or [f ~ v] for variation between [f] and [v], noting that a /u:/ ~ /ʊ/ contrast is maintained or lost, or indicating the change of a root in e.g. */'ni:l ~ 'nɛl+t/* for English *kneel ~ knelt*.

⟨∅⟩

A null segment or morpheme. This may indicate the absence of an affix, e.g. ⟨kæt-∅⟩ for where an affix might appear but does not (*cat* instead of *cats*), or a deleted segment that leaves a feature behind, such as ⟨∅^w⟩ for an theoretical labialized segment that is only realized as labialization on adjacent segments.^[104]

Capital letters

Full capital letters are not used as IPA symbols, except as typewriter substitutes (e.g. N for ⟨ŋ⟩, S for ⟨ʃ⟩, O for ⟨ɔ⟩ – see SAMPA). They are, however, often used in conjunction with the IPA in two cases:

1. for (archi)phonemes and for natural classes of sounds (that is, as wildcards). The extIPA chart, for example, uses capital letters as wildcards in its illustrations.
2. as carrying letters for the Voice Quality Symbols.

Wildcards are commonly used in phonology to summarize syllable or word shapes, or to show the evolution of classes of sounds. For example, the possible syllable shapes of Mandarin can be abstracted as ranging from /V/ (an atonic vowel) to /CGVN^T/ (a consonant-glide-vowel-nasal syllable with tone), and word-final devoicing may be schematized as C → C̥/_#. They are also used in historical linguistics for a sound that is posited but whose nature has not been determined beyond some generic category such as {nasal} or {uvular}. In speech pathology, capital letters represent indeterminate sounds, and may be superscripted to indicate they are weakly articulated: e.g. [P̥] is a weak indeterminate alveolar, [K̥] a weak indeterminate velar.^[114]

There is a degree of variation between authors as to the capital letters used, but these are ubiquitous in English-language material:^[115]

- ⟨C⟩ for {consonant}
- ⟨V⟩ for {vowel}
- ⟨N⟩ for {nasal}

Other common conventions are:^[116]

- ⟨T⟩ for {tone/accent} (tonicity)
- ⟨P⟩ for {plosive}
- ⟨F⟩ for {fricative}
- ⟨S⟩ for {sibilant}^[note 40]
- ⟨G⟩ for {glide/semivowel}
- ⟨L⟩ for {lateral} or {liquid}
- ⟨R⟩ for {rhotic} or {resonant/sonorant}^[note 41]
- ⟨C̥⟩ for {obstruent}
- ⟨□⟩ for {click}
- ⟨A, E, ɪ, O, U⟩ for {open, front, close, back, rounded vowel}^[note 42] and ⟨B, D, ʝ, K, Q, Φ, H⟩ for {labial, alveolar, post-alveolar/palatal, velar, uvular, pharyngeal, glottal^[note 43] consonant}, respectively
- ⟨X⟩ for {any sound}, as in ⟨CVX⟩ for a heavy syllable {CVC, CV̥, CV;}

The letters can be modified with IPA diacritics, for example:^[116]

- ⟨C'⟩ for {ejective}
- ⟨C̰⟩ for {implosive}
- ⟨N̠C⟩ or ⟨^NC⟩ for {prenasalized consonant}
- ⟨Ṽ⟩ for {nasal vowel}
- ⟨C^hV̥⟩ for {aspirated CV syllable with high tone}
- ⟨S̥⟩ for {voiced sibilant}
- ⟨N̥⟩ for {voiceless nasal}
- ⟨P͡F⟩ or ⟨P͡ɸ⟩ for {affricate}
- ⟨C^ɥ⟩ for a consonant with a glide as secondary articulation (e.g. ⟨Cʲ⟩ for {palatalized consonant} or ⟨C^w⟩ for {labialized consonant})
- ⟨D̪⟩ for {dental consonant}

⟨H⟩, ⟨M⟩, ⟨L⟩ are also commonly used for high, mid and low tone, with ⟨LH⟩ for rising tone and ⟨HL⟩ for falling tone, rather than transcribing them overly precisely with IPA tone letters or with ambiguous digits.^[note 44] When distinguishing five levels of pitch, ⟨xH⟩ and ⟨xL⟩ may be used for 'extra high' and 'extra low'. Arbitrary sequences of letters such as ⟨A B C D⟩ may be used for tone phonemes, especially when comparing across related languages.

Typical examples of archiphonemic use of capital letters are:

- ⟨I⟩ for the Turkish harmonic vowel set {i y u ʊ}^[note 45]
- ⟨D⟩ for the conflated flapped middle consonant of American English *writer* and *rider*
- ⟨N⟩ for the homorganic syllable-coda nasal of languages such as Spanish and Japanese (essentially equivalent to the wild-card usage of the letter)
- ⟨R⟩ in cases where a phonemic distinction between trill /r/ and flap /ɾ/ is conflated, as in Spanish *enrejar* /eNreˈxaR/ (the *n* is homorganic and the first *r* is a trill, but the second *r* is variable).^[117]

Similar usage is found for *phonemic* analysis, where a language does not distinguish sounds that have separate letters in the IPA. For instance, Castillian Spanish has been analyzed as having phonemes /θ/ and /s/, which surface as [θ] and [s] in voiceless environments and as [ð] and [z] in voiced environments (e.g. *hazte* /ˈaθte/ → [ˈaθte], vs *hazme* /ˈaθme/ → [ˈaðme], or *las manos* /laS ˈmanoS/ → [lazˈmanos]).^[118]

⟨V⟩, ⟨F⟩ and ⟨C⟩ have completely different meanings as Voice Quality Symbols, where they stand for "voice" (VoQS jargon for secondary articulation),^[note 46] "falsetto" and "creak". These three letters may take diacritics to indicate what kind of voice quality an utterance has, and may be used as carrier letters to extract a suprasegmental feature that occurs on all susceptible segments in a stretch of IPA. For instance, the transcription of Scottish Gaelic [k^{wh}u^{xw}t̪^wʃ^w] 'cat' and [k^{wh}u^{xw}t̪^fʃ^w] 'cats' (Islay dialect) can be made more economical by extracting the suprasegmental labialization of the words: V^w[k^hu^xt̪^ʃ] and V^w[k^hu^xt̪^fʃ].^[119] The conventional wildcards ⟨X⟩ or ⟨C⟩ might be used instead of VoQS ⟨V⟩ so that the reader does not misinterpret ⟨V^w⟩ as meaning that only vowels are labialized (i.e. X^w[k^hu^xt̪^ʃ] for all segments labialized, C^w[k^hu^xt̪^ʃ] for all consonants labialized), or the carrier letter may be omitted altogether (e.g. ^w[k^hu^xt̪^ʃ], [^wk^hu^xt̪^ʃ] or [k^hu^xt̪^ʃ]^w). (See § Suprasegmentals for other transcription conventions.)

This summary is to some extent valid internationally, but linguistic material written in other languages may have different associations with capital letters used as wildcards. For example, in German ⟨K⟩ and ⟨V⟩ are used for *Konsonant* 'consonant' and *Vokal* 'vowel'; in Russian, ⟨C⟩ and ⟨Г⟩ are used for согласный (*soglasnyj*, 'consonant') and гласный (*glasnyj*, 'vowel'). In French, tone may be transcribed with ⟨H⟩ and ⟨B⟩ for *haut* 'high' and *bas* 'low';^[120] Russian appears to be the opposite, with ⟨B⟩ for высокий (*vysokij*, 'high') and ⟨Н⟩ for низкий (*nizkij*, 'low').

vowel letter like *i* with a subscript omega (⟨ᵱ⟩) and unrounding a rounded letter like *u* with a subscript turned omega.^[124] As of 2024, the turned omega diacritic is in the pipeline for Unicode, and is under consideration for compression in extIPA.^[125] Kelly & Local use a combining *w* diacritic ⟨ᵱ⟩ for protrusion (e.g. ⟨y̥ᵱ⟩) and a combining *ɱ* diacritic ⟨ᵱ⟩ for compression (e.g. ⟨uᵱ⟩).^[126] Because their transcriptions are manuscript, these are effectively the same symbols as the old IPA diacritics, which indeed are historically cursive *w* and *ɱ*. However, the more angular ⟨ᵱ⟩ of typescript might misleadingly suggest the vowel is protruded and voiceless (like [ɱ]) rather than compressed and voiced.

Symbol names

In both print and speech, an IPA symbol is often distinguished from the sound it transcribes because IPA letters very often do not have their cardinal IPA values in practice. This is commonly the case in phonemic and broad phonetic transcription, making articulatory descriptions of IPA letters, such as "mid front rounded vowel" or "voiced velar stop", inappropriate as names for those letters. While the *Handbook of the International Phonetic Association* states that no official names exist for its symbols, it admits the presence of one or two common names for each.^[127] The symbols also have nonce names in the Unicode standard. In many cases, the names in Unicode and the IPA *Handbook* differ. For example, the *Handbook* calls ⟨ɛ⟩ "epsilon", while Unicode calls it "small letter open e".

The traditional names of the Latin and Greek letters are usually used for unmodified letters.^[note 50] Letters which are not directly derived from these alphabets, such as ⟨ɕ⟩, may have a variety of names, sometimes based on the appearance of the symbol or on the sound that it represents. In Unicode, some of the letters of Greek origin have Latin forms for use in IPA; the others use the characters from the Greek block.

For diacritics, there are two methods of naming. For traditional diacritics, the IPA notes the name in a well known language; for example, ⟨é⟩ is "e-acute", based on the name of the diacritic in English and French. Non-traditional diacritics are often named after objects they resemble, so ⟨ᵰ⟩ is called "d-bridge".

Geoffrey Pullum and William Ladusaw list a variety of names in use for both current and retired IPA symbols in their *Phonetic Symbol Guide*. Many of them found their way into Unicode.^[9]

Computer support

Unicode

Unicode supports nearly all of the IPA. Apart from basic Latin and Greek and general punctuation, the primary blocks are IPA Extensions, Spacing Modifier Letters and Combining Diacritical Marks, with lesser support from Phonetic Extensions, Phonetic Extensions Supplement, Combining Diacritical Marks Supplement, and scattered characters elsewhere. The extended IPA is supported primarily by those blocks and Latin Extended-G.

IPA numbers

After the Kiel Convention in 1989, most IPA symbols were assigned an identifying number to prevent confusion between similar characters during the printing of manuscripts. The codes were never much used and have been superseded by Unicode.

Many typefaces have support for IPA characters, but good diacritic rendering remains rare.^[129] Web browsers generally do not need any configuration to display IPA characters, provided that a typeface capable of doing so is available to the operating system.

Typefaces that provide full IPA and nearly full extIPA support, including properly rendering the diacritics, include Gentium, Charis SIL, Doulos SIL, and Andika developed by SIL International. Indeed, the IPA chose Doulos to publish their chart in Unicode format. In addition to the level of support found in commercial and system fonts, these fonts support the full range of old-style (pre-Kiel) staveless tone letters, through a character variant option that suppresses the stave of the Chao tone letters. They also have an option to maintain the [a] ~ [ɑ] vowel distinction in italics. The only notable gaps are with the extIPA: the combining parentheses, which enclose diacritics, are not supported, nor is the enclosing circle that marks unidentified sounds, and which Unicode considers to be a copy-edit mark and thus not eligible for Unicode support.

DejaVu is the second free Unicode font chosen by the IPA to publish their chart. It was last updated in 2016 and so does not support the Latin F or G blocks. Stacked diacritics tend to overstrike each other.

Proprietary system fonts

The Apple system fonts Geneva, Lucida Grande and Hiragino (certain weights) have only basic IPA support.

Brill has complete IPA and extIPA coverage of characters added to Unicode by 2020, with good diacritic and tone-letter support. It is a commercial font but is freely available for non-commercial use.^[131]

Several systems have been developed that map the IPA symbols to ASCII characters. Notable systems include SAMPA and X-SAMPA. The usage of mapping systems in on-line text has to some extent been adopted in the context input methods, allowing convenient keying of IPA characters that would be otherwise unavailable on standard keyboard layouts.

The sequence □-H□-H□-k□[˘]□ in the fonts Gentium, Andika, Brill, Noto Serif, DejaVu Sans and Calibri. All but DejaVu align diacritics well. Asterisks are characters not supported by that font. In Noto and DejaVu, the red tone letters do not link properly, and in DejaVu the stacked diacritics overstrike each other. This is a test sequence: Noto and Calibri support most IPA adequately.

IETF language tags

IETF language tags have registered `fonipa` as a variant subtag identifying text as written in IPA.^[132] Thus, an IPA transcription of English could be tagged as `en-fonipa`. For the use of IPA without attribution to a concrete language, `und-fonipa` is available.

Computer input using on-screen keyboard

Online IPA keyboard utilities are available, though none of them cover the complete range of IPA symbols and diacritics. Examples are the *IPA 2018 i-charts* hosted by the IPA,^[133] *IPA character picker* by Richard Ishida at GitHub,^[134] *Type IPA phonetic symbols* at TypeIt.org,^[135] and an *IPA Chart keyboard* by Weston Ruter also at GitHub.^[136] In April 2019, Google's Gboard for Android added an IPA keyboard to its platform.^{[137][138]} For iOS there are multiple free keyboard layouts available, such as the *IPA Phonetic Keyboard*.^[139]

See also

- Afroasiatic phonetic notation – Phonetic notation
- Americanist phonetic notation – Phonetic alphabet developed in the 1880s
- Articulatory phonetics – Branch of linguistics studying how humans make sounds
- Case variants of IPA letters – International Phonetic Alphabet variants
- Cursive forms of the International Phonetic Alphabet – Deprecated cursive forms of IPA symbols
- Extensions to the International Phonetic Alphabet – Disordered speech additions to the phonetic alphabet
- Index of phonetics articles
- International Alphabet of Sanskrit Transliteration – Transliteration scheme for Indic scripts
- Sound correspondences between English accents
- List of international common standards
- Luciano Canepari – Italian linguist (born 1947)
- Phonetic symbols in Unicode
- RFE Phonetic Alphabet
- SAMPA – Computer-readable phonetic script
- Semyon Novgorodov – Yakut politician and linguist – inventor of IPA-based Yakut scripts
- TIPA – TeX macro package provides IPA support for LaTeX
- UAI phonetic alphabet – Phonetic transcription
- Uralic Phonetic Alphabet – Phonetic alphabet for Uralic languages
- Voice Quality Symbols – Set of phonetic symbols used for voice quality, such as to transcribe disordered speech
- X-SAMPA – Remapping of the IPA into ASCII

Notes

1. The inverted bridge minus under the ⟨t̪^h⟩ specifies it as apical (pronounced with the tip of the tongue), and the superscript *h* shows that it is aspirated (breathy); the rectangle under ⟨t̪⟩ specifies that it is laminal (pronounced with the blade of the tongue). These details cause the English /t/ to sound different from the French /t/.
2. "Originally, the aim was to make available a set of phonetic symbols which would be given *different* articulatory values, if necessary, in different languages."^[7]

3. "From its earliest days [...] the International Phonetic Association has aimed to provide 'a separate sign for each distinctive sound; that is, for each sound which, being used instead of another, in the same language, can change the meaning of a word' [...] what became widely known in the twentieth century as the phoneme."^[12]
4. exceptions are affricates and diphthongs, which may be written as either simple sequences, such as ⟨ts⟩ and ⟨au⟩, or with tie bars or diacritics such as ⟨ts̺⟩ and ⟨au̯⟩ to clarify that they are single sounds.
5. For instance, flaps and taps are two different kinds of articulation, but since no language has (yet) been found to make a distinction between, say, an alveolar flap and an alveolar tap, the IPA does not provide such sounds with dedicated letters. Instead, it provides a single letter – in this case, [ɾ] – for both. Strictly speaking, this makes the IPA a partially phonemic alphabet, not a purely phonetic one.
6. This exception to the rules was made primarily to explain why the IPA does not make a dental–alveolar distinction, despite one being phonemic in hundreds of languages, including most of the continent of Australia. Americanist Phonetic Notation makes (or at least made) a distinction between apical ⟨t d s z n l⟩ and laminal ⟨ɾ ɸ ʒ ʒ v λ⟩, which is easily applicable to alveolar vs dental (when a language distinguishes apical alveolar from laminal dental, as in Australia), but despite several proposals to the Council, the IPA never voted to accept such a distinction. There are however other common phonemic distinctions that are made with diacritics, such as ⟨ʰ⟩ for aspirated consonants – a phonemic distinction made by thousands of languages – and ⟨ɹ̺⟩ for palatalized consonants, which were once transcribed by distinct letters in the IPA until those letters were retired in favor of the diacritic.
7. There are three basic tone diacritics and five basic tone letters, both sets of which may be compounded.
8. "The non-roman letters of the International Phonetic Alphabet have been designed as far as possible to harmonize well with the roman letters. The Association does not recognize makeshift letters; It recognizes only letters which have been carefully cut so as to be in harmony with the other letters."^[14]
9. "The new letters should be suggestive of the sounds they represent, by their resemblance to the old ones."^[15]
10. Originally, [ɥ] was written as a small capital U. However, this was not easy to read, and so it was replaced with a turned small capital omega. In modern typefaces, it often has its own design, called a "horseshoe".
11. For example, Merriam-Webster dictionaries use backslashes \ ... \ to demarcate their in-house diaphonemic transcription system. This contrasts with the *Oxford English Dictionary*, which transcribes a specific target accent.
12. For example, single and double pipe symbols are used for minor and major prosodic breaks. Although the *Handbook* specifies the prosodic symbols as being "thick" vertical lines, which would in theory be distinct from simple ASCII pipes used as delimiters (and similar to *Dania* transcription), this was an idea to keep them distinct from the otherwise similar pipes used as click letters, and is almost never found in practice.^[27] The *Handbook* assigns the prosodic pipe the Unicode encodings U+007C, which is the simple ASCII symbol, and the double pipe U+2016.^[28]
13. The proper angle brackets in Unicode are the mathematical symbols (U+27E8 and U+27E9). Chevrons ⟨...⟩ (U+2039, U+203A) are sometimes substituted, as in Americanist phonetic notation, as are the less-than and greater-than signs <...> (U+003C, U+003E) found on ASCII keyboards.
14. *Pronunciation respelling for English* contains detailed comparisons of respelling systems.
15. Monolingual Hebrew dictionaries use pronunciation respelling for words with unusual spelling; for example, the *Even-Shoshan Dictionary* respells תְּכַנִּית as ⟨תְּכַנִּית⟩ because the word uses the *kamatz katan*.
16. For example, Sergey Ozhegov's dictionary adds [нэ́] in brackets to the French loan-word пеще (pince-nez) to indicate that the final ⟨e⟩ does not iotate the preceding ⟨н⟩.
17. "In accordance with long-established Czech lexicographical tradition, a modified version of the International Phonetic Alphabet (IPA) is adopted in which letters of the Czech alphabet are employed."^[52]
18. "Segments can usefully be divided into two major categories, consonants and vowels."^[59]
19. They were moved "for presentational convenience [...] because of [their] rarity and the small number of types of sounds which are found there."^[62]
20. "A symbol such as [β], shown on the chart in the position for a voiced bilabial fricative, can also be used to represent a voiced bilabial approximant if needed."^[65]

21. It is traditional to place the tie bar above the letters. It may be placed below to avoid overlap with ascenders or diacritic marks, or simply because it is more legible that way, as in Niesler; Louw; Roux (2005). "Phonetic analysis of Afrikaans, English, Xhosa and Zulu using South African speech databases".^[70]
22. Cf. the /^h.../ and /^l.../ transcriptions in Ernst-Kurdi, Eszter (2017). "The Phonology of Mada" (<https://www.sil.org/resources/archives/76424>). SIL Yaoundé.
23. E.g. Dolgopolsky, Aaron (2013). *Indo-European Dictionary with Nostratic Etymologies* (<https://searchworks.stanford.edu/view/11657956>). Studia Philologica. Rukopisnye pamiatniki Drevnei Rusi.
24. The IPA *Handbook* variously defines the "linking" symbol as marking the "lack of a boundary"^[78] or "absence of a break",^[28] and gives French *liaison* and English *linking r* as examples. It more generally means that the consonant ending one word forms a syllable with the vowel beginning the following word, across an orthographic space. However, the *Handbook* illustration for Croatian uses it to tie atonic *clitics* to tonic words, with no resulting change in implied syllable structure.
25. The global rise and fall arrows come before the affected syllable or prosodic unit, like stress and upstep/downstep. This contrasts with the Chao tone letters (listed below), which most commonly come after. One will occasionally see a horizontal arrow ⟨→⟩ for global level pitch (only dropping due to downdrift), e.g. in Julie Barbour (2012) *A Grammar of Neverver*. Additionally, some fonts display the arrows as emoji by default, if ︎ is not appended.
26. There is not a one-to-one correspondence between tone/pitch diacritics and tone/pitch letters. When pitch is transcribed with diacritics, the three pitches ⟨é ē è⟩ are taken as the basic levels and are called 'high', 'mid' and 'low'. Contour tones combine only these three and are called ⟨ê⟩ 'high-mid' etc. The more extreme pitches, which do not form contours, are ⟨ë⟩ 'extra-high' and ⟨è⟩ 'extra-low', using doubled diacritics. When transcribed with tone letters, however, combinations of all five levels are possible. Thus, ⟨eᵀ eᵀ eᵀ⟩ may be called 'high', 'mid' and 'low', with ⟨eᵀ eᵀ⟩ being 'near-high' and 'near-low', analogous to descriptions of vowel height. In a three-level transcription, ⟨é ē è⟩ are identified with ⟨eᵀ eᵀ eᵀ⟩, but in a five-level transcription, ⟨ë è⟩ are identified with ⟨eᵀ eᵀ⟩.^[79]
27. The peaking and dipping diacritics are uncommon and are not illustrated in the IPA *Handbook*. An example is /klâ/ 'properly' in Mian.
28. Although any combination of tone diacritics is theoretically possible, such as ⟨ễ⟩ for a falling–rising–falling tone, any others than those illustrated are rare. Double acute – grave is found for an extra-high falling tone in Valentin Vydrin, 2020, *Dan*. In Vossen & Dimmendaal (eds.) *The Oxford Handbook of African Languages*; and double acute – macron is found for an extrahigh-to-mid tone in Roger Blench, 2025, *Tone Systems in the Bantoid Languages*. Neither is supported by Unicode.
29. For example, "Balearic" (<https://www.merriam-webster.com/dictionary/Balearic>). *Merriam-Webster.com Dictionary*. Merriam-Webster.
30. Russian and Lithuanian sources and commonly use the character U+2E3D ◌͡ VERTICAL SIX DOTS for a less-than-minor break, such as the slight break in list intonation (e.g. the very slight break between digits in a telephone number). U+2E3E ◌͢ WIGGLY VERTICAL LINE is used for an unexpected interruption in or a sharp change of intonation.^[82]
31. A work-around sometimes seen when a language has more than one rising or falling tone, and the author wishes to avoid the poorly legible diacritics ⟨ẽ, ẽ̃, ê, è⟩ but does not wish to employ tone letters, is to restrict the generic rising ⟨ẽ̃⟩ and falling ⟨ễ⟩ diacritics to the higher-pitched of the rising and falling tones, say /eᵀᵀ/ and /eᵀᵀ/, and to resurrect the retired (pre-Kiel) IPA subscript diacritics ⟨eᵀ⟩ and ⟨eᵀ⟩ for the lower-pitched rising and falling tones, say /eᵀᵀ/ and /eᵀᵀ/. When a language has either four or six level tones, the two middle tones are sometimes transcribed as high-mid ⟨ễ⟩ (non-standard) and low-mid ⟨ễ̃⟩. Non-standard ⟨ễ̃⟩ is occasionally seen combined with acute and grave diacritics or with the macron to distinguish contour tones that involve the higher of the two mid tone levels.
32. See for example Pe Maung Tin (1924). "b³·mizː". *Le Maître Phonétique*. 2 (39) (5): 4–5. JSTOR 44704085 (<https://www.jstor.org/stable/44704085>). where five pitch levels are distinguished.
33. The example has changed over the years. In the chart included in the 1999 IPA *Handbook*, it was [ᵀᵀ], and since the 2018 revision of the chart it has been [ᵀᵀ].
34. Chao did not include tone shapes such as [ᵀᵀ], [ᵀᵀ], which rise or fall and then level off (or vice versa). Such tone shapes are, however, frequently encountered in the modern literature.

35. In Chao's Sinological convention, a single tone letter ⟨ṽ⟩ is used for a high tone on a checked syllable, and a double tone letter ⟨ṽṽ⟩ for a high tone on an open syllable. Such redundant doubling is not used in the *Handbook*, where the tones of Cantonese [siṽ] 'silk' and [sɪkṽ] 'color' are transcribed the same way. If the author wishes to indicate a difference in phonetic or phonemic length, the IPA accomplishes that with the length marks ⟨◌ː◌ː◌ː⟩ rather than through the tone letters.
36. Sometimes the obsolete transcription ⟨kʰ⟩ (with a turned apostrophe) for weak aspiration vs. ⟨kʰ⟩ for strong aspiration is still seen.
37. E.g. in Laver 1994, pp. 559–560
38. The motivation for this may vary. Some authors find the tie bars displeasing but the lack of tie bars confusing (i.e. ⟨č̌⟩ for /t͡ʃ/ as distinct from /tʃ/), while others simply prefer to have one letter for each segmental phoneme in a language.
39. "At the 1989 Kiel Convention of the IPA, a sub-group was established to draw up recommendations for the transcription of disordered speech."^[108]
40. As in Afrasianist phonetic notation. ⟨S⟩ is particularly ambiguous. It has been used for 'stop', 'fricative', 'sibilant', 'sonorant' and 'semivowel'. On the other hand, plosive/stop is frequently abbreviated ⟨P⟩, ⟨S⟩ or (with non-tonal languages) * ⟨T⟩. The illustrations given here use, as much as possible, letters that are capital versions of members of the sets they stand for: IPA [n] is a nasal and ⟨N⟩ is any nasal; [p] is a plosive, [f] a fricative, [s] a sibilant, [l] both a lateral and a liquid, [r] both a rhotic and a resonant, and [ɣ] a click. ⟨ϕ⟩ is an obstruent in Americanist notation, where it stands for [ts]. Alternative wildcards for 'glide', ⟨J⟩ and ⟨W⟩, partly fit this pattern, but are much less common than ⟨G⟩ in English-language sources.
41. In the context of ⟨CRV-⟩ syllables, the ⟨R⟩ is understood to include liquids and glides but to exclude nasals, as in Bennett (2020: 115) 'Click Phonology', in Sands (ed.), *Click Consonants*, Brill
42. {Close vowel} may instead be ⟨U⟩, and ⟨O⟩ may stand for {obstruent}.
43. Or glottal~pharyngeal ⟨H⟩, as in Afrasianist phonetic notation.
44. Somewhat more precisely, ⟨LM⟩ and ⟨MH⟩ are sometimes used for low and high rising tones, and ⟨HM⟩, ⟨ML⟩ for high and low falling tones; occasionally ⟨R⟩ for 'rising' or ⟨F⟩ for 'falling' is seen.
45. For other Turkic languages, ⟨I⟩ may be restricted to {u i} (that is, to *ı* *î*), ⟨U⟩ to *u* *ü*, ⟨A⟩ to *a* *e* (or *a* *ä*), etc.
46. VoQS ⟨V⟩ does not mean phonetic voicing, nor a vowel; for example, in VoQS ⟨Ṽ⟩ is "nasal voice" (that is, nasalization), not a nasal vowel as it would be read in IPA notation.
47. "Diacritics may also be employed to create symbols for phonemes, thus reducing the need to create new letter shapes."^[121]
48. Dedicated letters have been proposed, such as rotated ⟨Ḅ⟩ and ⟨Ḟ⟩, reversed ⟨Ḥ⟩ and ⟨Ḭ⟩, or small-capital ⟨Ḃ⟩ and ⟨Ḑ⟩. Ball, Rahilly & Lowry (2017) *Phonetics for speech pathology*, 3rd edition, Equinox, Sheffield.
49. "The diacritics...can be used to modify the lip or tongue position implied by a vowel symbol."^[123]
50. For example, the IPA *Handbook* lists ⟨p̥⟩ as "lower-case P" and ⟨χ̥⟩ as "chi."^[128]

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External links

- Official website (<http://www.internationalphoneticassociation.org>)
 - Interactive IPA chart (https://www.internationalphoneticassociation.org/IPAcharts/inter_chart_2018/IPA_2018.html)
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